

HARDEEP SINGH

AI Engineer | New Delhi, India | hardeepsindia@gmail.com | +91 97175 17181 | github.com/rav4nn

PROFILE

Self-taught AI engineer with 200+ organic users across deployed products and 60+ GitHub stars on open-source tools. Builds and ships end-to-end LLM-powered applications using Python, FastAPI, Next.js, and the full AI stack - RAG pipelines, vector search, prompt engineering, and multi-provider LLM integration (Anthropic, OpenAI, Gemini, DeepSeek). Co-built a crisis-response platform serving 15 Indian states in 48 hours.

PROJECTS

Coffee Coach coffeecoach.app

Python · FastAPI · Next.js · React · PostgreSQL · Docker · LLM APIs

- Full-stack AI coaching app for specialty coffee brewers — personalised feedback from an AI coach (Kapi) based on brew history and taste data.
- 64 real users acquired organically via Twitter/X and Reddit with zero paid promotion.
- End-to-end production deployment: Next.js on Vercel, FastAPI on Hetzner VPS with Docker, Nginx, and Certbot.

YouTube RAG Scraper github.com/rav4nn/youtube-rag-scraper

Python · LangChain · FAISS · RAG · Embeddings

- Pipeline that scrapes YouTube channels, extracts transcripts, embeds them into a FAISS vector store, and exposes a semantic search interface over the resulting knowledge base.
- 60 GitHub stars and 13 forks gained organically within a month — most-starred project in the portfolio.
- Full RAG stack: ingestion, chunking, embedding, retrieval, and generation.

buildinpublic-x github.com/rav4nn/buildinpublic-x

Python · GitHub Actions · LLM APIs · Multi-provider (Anthropic, OpenAI, Gemini, Groq)

- GitHub Action that reads commit history and README, generates build-in-public threads via LLM, and auto-posts to X and Bluesky on a cron schedule.
- Zero backend or database required - runs entirely inside GitHub Actions. LLM provider swappable via a single environment variable.
- Costs ~\$0.02 per post. Designed for developers who build in public without the overhead of manual writing.

Flux RAG github.com/rav4nn/flux-rag

Python · RAG · Vector Search · Hybrid Retrieval · Re-ranking

- Advanced RAG research project covering chunking strategies, hybrid search, re-ranking, and retrieval evaluation — built to develop rigorous, production-applicable understanding of the full retrieval stack.

Splitwala - WhatsApp Expense Splitter github.com/rav4nn/splitwala-webjs

Node.js · whatsapp-web.js

- WhatsApp bot for group expense splitting via simple commands (/split, /summary, balance tracking). Live on a real number with active users.
- Demonstrates cross-language comfort beyond Python and focus on zero-friction UX.

EXPERIENCE

Operations Lead - CovidWin Apr 2021 – May 2021

Volunteer Crisis Platform · Python · Google Sheets API · Cron Automation · covidwin.in

- Co-built a COVID-19 resource platform from zero to full operations in 48 hours during India's second wave — covering 15 states, 50 cities, and 10,000 verified life-saving resources.
- Led volunteer coordination across 8 states, managing 4,000 volunteers from partner organisations.
- Contributed to a 15-minute automated data sync pipeline using Google Sheets API with deduplication and multi-source aggregation.

SKILLS

Languages	Python, JavaScript (Node.js), HTML/CSS
AI / ML	LangChain, FAISS, RAG Pipelines, Prompt Engineering, Embeddings, Vector Search, Re-ranking
LLM APIs	Anthropic (Claude), OpenAI, DeepSeek, Gemini, Groq
Backend	FastAPI, REST APIs, PostgreSQL, Next.js, React, TailwindCSS, Docker, Nginx
Deployment	Hetzner VPS, Vercel, Railway, Fly.io, Cloudflare Pages, Git/GitHub, GitHub Actions

EDUCATION

Chemical Engineering	2018
Indian Institute of Technology (IIT), Delhi	
Class XII - CBSE	2013
St. Mary's School, Delhi	